

Topology Assignment

1. Show that if $\mathcal{A}$ is a basis for a topology on X , then the topology generated by $\mathcal{A}$ equals the intersection of all topologies on X that contain $\mathcal{A}$
2. Consider the following collection of subsets of $\mathbf{R}$.

$$\mathcal{B}_1 = \{(a, b) | a < b\}$$

$$\mathcal{B}_2 = \{[a, b) | a < b\}$$

$$\mathcal{B}_3 = \{(a, b] | a < b\}$$

$$\mathcal{B}_4 = \mathcal{B}_1 \cup \{B - K | B \in \mathcal{B}_1, K = \{1/n | n \in \mathbf{N}\}\}$$

$$\mathcal{B}_5 = \{(a, +\infty) | a \in \mathbf{R}\}$$

$$\mathcal{B}_6 = \{(-\infty, a) | a \in \mathbf{R}\}$$

$$\mathcal{B}_7 = \{B | \mathbf{R} - B \text{ is finite}\}$$
 - (a) Show that each $\mathcal{B}_i$, is a basis for a topology on $\mathbf{R}$.
 - (b) Compare these seven topologies with one another.
 - (c) Show that $\mathcal{B}_5 \cup \mathcal{B}_6$, is a subbasis that generates the same topology as $\mathcal{B}_1$
3. (a) Apply Lemma 2.3 to show that the countable collection $\mathcal{B}_1' = \{(a, b) | a < b \in \mathbf{Q}\}$ is a basis that generates the standard topology on $\mathbf{R}$.
 (b) Show that the collection $\mathcal{B}_2' = \{[a, b) | a < b, a, b \in \mathbf{Q}\}$ is a basis that generates a topology different from the lower limit topology on $\mathbf{R}$.
4. Show that if Y is a subspace of X , and A is a subset of Y , then the subspace topology on A as a subspace of Y is the same as the subspace topology on A as a subspace of X .
5. A map $f : X \rightarrow Y$ is said to be an open map if for every open set U of X , the set $f(U)$ is open in Y . Show that $\Pi_1 : X \times Y \rightarrow X$ and $\Pi_2 : X \times Y \rightarrow Y$ are open maps.
6. (a) Show that the collection of open rays in an ordered set A is a sub-basis for the order topology on A .
 (b) Let X be an ordered set in the order topology. Let Y be an interval or ray in X ; let $(-\infty, a)$ and $(a, +\infty)$ be open rays in X . Show that if $a \in Y$, then each of the sets

- $(-\infty, a) \cap Y$ and $(a, +\infty) \cap Y$ is an open ray of the ordered set Y , while if $a \notin Y$, each of these sets is either empty or all of Y
- (c) Conclude that if Y is an interval or ray in X , then the order to the subspace topology on Y are the same.
7. Let I denote the subspace $[0, 1]$ of $\mathbf{R}$. Compare the product topology on $I \times I$, the dictionary order topology on $I \times I$, and the topology $I_d \times I$, where I_d denotes I in the discrete topology.
 8. Show that if A is closed in Y and Y is closed in X , then A is closed in X .
 9. Show that if A is closed in X and B is closed in Y , then $A \times B$ is closed in $X \times Y$.
 10. Let A , B , and A_α denote subsets of a space X . Prove the following:
 - (a) $\overline{A \cup B} = \overline{A} \cup \overline{B}$
 - (b) $\overline{\cup A_\alpha} \supset \cup \overline{A_\alpha}$; give an example where the equality fails.
 11. Show that every order topology is Hausdorff.
 12. $A \subset X$, we define the boundary of A by the equation: $\text{Bd}A = \overline{A} \cap (\overline{X - A})$
 - (a) Show that $\text{Int } A$ and $\text{Bd } A$ are disjoint, and $\overline{A} = \text{Int } A \cup \text{Bd } A$.
 - (b) Show that $\text{Bd}A = \emptyset \iff A$ is both open and closed
 - (c) Show that U is open $\iff \text{Bd}U = \overline{U} - U$
 - (d) if U is open. Is it true that $U = \text{Int}(\overline{U})$? Justify your answer
 13. Find the boundary and the interior of each of the following subsets of $\mathbf{R}^2$
 - (a) $A = \{x \times y | y = 0\}$
 - (b) $B = \{x \times y | x > 0 \text{ and } y \neq 0\}$
 - (c) $C = A \cup B$
 - (d) $D = \{x \times y | x \in \mathbf{Q}\}$
 - (e) $E = \{x \times y | 0 < x^2 + y^2 \leq 1\}$
 - (f) $F = \{x \times y | x \neq 0 \text{ and } y \leq 1/x\}$
 14. Let $A \subset X$. Show that if C is a connected subset of X that intersects both A and $X - A$, then C intersects $\text{Bd}A$.
 15. Let $Y \subset X$; let X and Y be connected. Show that if A and B form a separation of $X - Y$, then $Y \cup A$ and $Y \cup B$ are connected.
 16. a) Show that no two of the spaces $(0, 1)$, $(0, 1]$, and $[0, 1]$ are homeomorphic.
 (b) Suppose that $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are imbeddings. Show by means of an example that X and Y need not be homeomorphic.
 17. Let $f : X \rightarrow X$ be continuous. Show that if $X = [0, 1]$ there is a point x such that $f(x) = x$. The point x is called a fixed point. What happens if X equals $[0, 1)$ or $(0, 1)$?
 18. a) Is a product of path-connected spaces necessarily path connected?
 (b) If $A \subset X$ and A is path connected, is $\overline{A}$ necessarily path connected?
 (c) If $f : X \rightarrow Y$ is continuous and X is path connected, is $f(X)$ necessarily path connected?

- (d) If $\{A_\alpha\}$ is a collection of path-connected subsets of X and if $\cap A_\alpha = \emptyset$, is $\cup A_\alpha$, necessarily path connected ?
19. If $f : X \rightarrow Y$ is continuous and X is locally connected, is $f(X)$ necessarily locally connected? What if f is both continuous and open?
20. Show that $I \times I$ in the dictionary order topology is locally connected but not locally path connected. What are the path components of this space?
21. Show that a finite union of compact sets is compact.
22. Let A and B be disjoint compact subsets of the Hausdorff space X . Show that there exist disjoint open sets U and V containing A and B , respectively.
23. Theorem. Let $f : X \rightarrow Y$; let Y be compact Hausdorff. Then f is continuous if and only if the graph of f .

$$G_f : \{x \times f(x) | x \in X\}$$

, is closed in $X \times Y$. [Hint: If G_f is closed and V is a neighborhood of $f(x_0)$, find a tube about $x_0 \times (Y - V)$ not intersecting G_f .]

24. Generalize the tube lemma as follows:

Theorem. Let A and B be subsets of X ; and Y , respectively; let N be an open set in $X \times Y$ containing $A \times B$. If B is compact, then there exists an open set U in X such that

$$A \times B \subset U \times B \subset N$$

. If A and B are both compact, then there exist open sets U and V in X and Y , respectively, such that

$$A \times B \subset U \times V \subset N$$

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25. Let X be a metric space with metric d ; let A be a subset of X . Recall that the diameter of A is defined by the equation $\text{diam } A = \text{lub } \{d(a_1, a_2) | a_1, a_2 \in A\}$, if it exists. Show that if A is compact, then $\text{diam } A$ exists and equals $d(a_1, a_2)$ for some $a_1, a_2 \in A$.
26. Show that the rationals $\mathbf{Q}$ are not locally compact.
27. Show that the conditions in Theorem 8.1 characterize the one-point compactification up to homeomorphism
28. Show that if X has a countable basis, every collection of disjoint open X is countable.
29. Let X have a countable basis; let A be an uncountable subset of X .
- (a) Show that the subspace topology on A is not the discrete topology.
- (b) Show that A contains at least one of its limit points.

- (c) Show that A contains uncountably many of its limit points.
30. Show that if X has a countable basis $\{B_n\}$, then every basis $\mathcal{C}$ for X contains a countable basis for X . [Hint: For every pair of indices n, m for which it is possible, choose $C_{n,m} \in \mathcal{C}$ such that $B_n \subset C_{n,m} \subset B_m$. Let $\mathcal{C}'$ be the collection of the sets $C_{n,m}$]
 31. Show that $\mathbf{R}_I$ and $I \times I$ in the dictionary order topology are not metrizable
 32. Show that if X is regular, every pair of points of X have neighborhoods whose closures are disjoint.
 33. Show that if X is normal, every pair of disjoint closed sets have neighborhoods whose closures are disjoint.
 34. Show that a closed subspace of a normal space is normal.
 35. Show that every order topology is regular
 36. (a) Show that a connected normal space having more than one point is uncountable.
(b) Show that a connected regular space having more than one point is uncountable. [Hint: Any countable space is Lindelöf.]
 37. Let X be a regular space with a countable basis; let U be open in X .
(a) Show that U equals a countable union of closed sets of X .
(b) Show there is a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) > 0$ for $x \in U$ and $f(x) = 0$ for $x \notin U$
 38. Let Z be a topological space. If Y is a subspace of Z , we say that Y is a retract of Z if there is a continuous map $\gamma : Z \rightarrow Y$ such that $\gamma(y) = y$ for each $y \in Y$
(a) Show that if Z is Hausdorff and Y is a retract of Z , then Y is closed in Z .
(b) Let A be a two-point set in $\mathbf{R}^2$. Show that A is not a retract of $\mathbf{R}^2$.
(c) Let S be the unit circle in $\mathbf{R}^2$; show that S^1 is a retract of $\mathbf{R}^2 - \{0\}$, where $\{0\}$ is the origin. Can you conjecture whether or not S^1 is a retract of $\mathbf{R}^2$
 39. Let $X_1 \subset X_2 \subset \dots$ be a sequence of spaces, where X_i is a closed subspace of X_{i+1} for each i . Let $X = \cup X_i$. Suppose that we define a subset U of X to be open in X if $U \cap X_i$ is open in X_i for each i . The topology we obtain is called the topology on X coherent with the spaces X_i .
(a) Show that X_i is a subspace of X in this topology.
(b) Show that if each space X_i is normal, so is the space X . [Hint: Use the Tietze theorem].